

HUIJIE ZHANG

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EDUCATION

University of Michigan, Ann Arbor <i>Ph.D. in Electrical Engineering and Computer Science</i>	Ann Arbor, U.S.A Sep. 2023 - May. 2027 (expected)
• Advisor: Prof. Qing Qu • Areas of Study: Diffusion Model, Generalization, Interpretability • GPA: 4.0/4.0	
<i>Research Assistant in Electrical Engineering and Computer Science</i>	Dec. 2022 - May. 2023
• Advisor: Prof. Qing Qu • Areas of Study: Diffusion Model	
<i>M.S.E in Mechanical Engineering</i>	Sep. 2020 - Dec. 2022
• Advisor: Prof. Odest Chadwicke Jenkins • Areas of Study: Robot Perception, Transparent Objects Perception • GPA: 4.0/4.0	
Huazhong University of Science and Technology <i>B.S. in Mechanical Engineering</i>	Wuhan, China Sep. 2016 - May. 2020
• Advisor: Prof. Zhigang Wu • Areas of Study: Soft Robot • GPA: 3.9/4.0	

WORKING EXPERIENCE

DeepMind, Google <i>Student Researcher</i>	San Francisco, U.S.A Jun. 2026 - Dec. 2026 (expected)
• Mentor: Joe Ortiz and Ruiqi Gao • Exploring diffusion forcing for long-video generation.	
Creative vision team, Snap Inc. <i>Research Intern</i>	Los Angeles, U.S.A Jun. 2025 - Dec. 2025
• Mentor: Ivan Skorokhodov, Aliaksandr Siarohin and Sergey Tulyakov • Developed a few-step diffusion model for efficient generative modeling. Proposed AlphaFlow, achieving state-of-the-art performance on the ImageNet benchmark.	

HONORS AND FELLOWSHIPS

2026	ICML 2026 Gold Reviewers	
2026	Rackham Predoctoral Fellowship	[Website]
2025	NeurIPS 2025 Outstanding Reviewers	[Website]
2025	Publications reported by SIAM	[Website]
2025	ICLR 2025 Outstanding Reviewers,	[Website]
2024	NeurIPS 2024 Outstanding Reviewers,	[Website]
2023	NeurIPS workshop Best Paper award,	[Website] / [News from Umich]
2023	Ph.D. Student Scholarship, University of Michigan, Ann Arbor	

PUBLICATIONS

PREPRINTS

1. **Huijie Zhang**, Zijiang Huang, Siyi Chen, Jinfan Zhou, Zekai Zhang, Peng Wang and Qing Qu. "Understanding Generalization in Diffusion Models via Probability Flow Distance." [\[Arxiv\]](#)

PEER-REVIEWED CONFERENCE PROCEEDINGS AND JOURNAL ARTICLES

1. Peng Wang*, **Huijie Zhang***, Zekai Zhang, Siyi Chen, Yi Ma, and Qing Qu. "Breaking the Curse of Dimensionality: Diffusion Models Efficiently Learn Low-Dimensional Distributions." *The Journal of Machine Learning Research (JMLR 2026)*. [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)
2. **Huijie Zhang**, Aliaksandr Siarohin, Willi Menapace, Michael Vasilkovsky, Sergey Tulyakov, Qing Qu and Ivan Skorokhodov. "AlphaFlow: Understanding and Improving MeanFlow Models." *The Fourteenth International Conference on Learning Representations (ICLR 2026)* [\[Arxiv\]](#)/[\[Code\]](#)
3. **Huijie Zhang***, Jinfan Zhou*, Yifu Lu, Minzhe Guo, Peng Wang, Liyue Shen, and Qing Qu. "The emergence of reproducibility and consistency in diffusion models." *The Forty-first International Conference on Machine Learning (ICML 2024)* [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)/[\[News\]](#)/[\[Talk\]](#)
🏆 **Best paper award**, [NeurIPS Diffusion Model Workshop 2023](#)
4. **Huijie Zhang***, Yifu Lu*, Ismail Alkhouri, Saiprasad Ravishankar, Dogyoon Song, and Qing Qu. "Improving Training Efficiency of Diffusion Models via Multi-Stage Framework and Tailored Multi-Decoder Architecture." *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*. [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)
5. **Huijie Zhang**, Peng Wang, Siyi Chen, Zekai Zhang, and Qing Qu. "Generalization of diffusion models: Principles, theory, and implications." *SIAM News (2025)*. [\[Website\]](#)
6. Wenda Li*, **Huijie Zhang***, and Qing Qu. "Shallow Diffuse: Robust and Invisible Watermarking through Low-Dimensional Subspaces in Diffusion Models." *The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*. (**Spotlight, top 3.2%**) [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)
7. Siyi Chen*, **Huijie Zhang***, Minzhe Guo, Yifu Lu, Peng Wang, and Qing Qu. "Exploring low-dimensional subspaces in diffusion models for controllable image editing." *The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)* [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)
8. Lianghe Shi*, Meng Wu*, **Huijie Zhang**, Zekai Zhang, Molei Tao and Qing Qu. "A Closer Look at Model Collapse: From a Generalization-to-Memorization Perspective." *The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*. (**Spotlight, top 3.2%**) [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)
🏆 **Oral Presentation**, [ICML MemFM Workshop 2025](#)
9. Xiaotong Chen, **Huijie Zhang**, Zeren Yu, Anthony Opipari, and Odest Chadwicke Jenkins. "Clearpose: Large-scale transparent object dataset and benchmark." *The European conference on computer vision*, pp. 381-396. (ECCV 2022) [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)
10. Xiaotong Chen, **Huijie Zhang**, Zeren Yu, Stanley Lewis, and Odest Chadwicke Jenkins. "Progresslabeller: Visual data stream annotation for training object-centric 3d perception." *The 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 13066-13073. (IROS 2022) [\[Arxiv\]](#)/[\[Code\]](#)/[\[Website\]](#)

TEACHING

University of Michigan, Ann Arbor

Graduate Student Instructor for [EECS 453](#)

Ann Arbor, U.S.A

Sep. 2024 - Dec. 2024

- Developed course materials, led discussions, and provided student support through office hours for a course on machine learning.
- Teaching Evaluation: 4.7/5.0

University of Michigan, Ann Arbor

High School Summer Camp Instructor for [AI Magic](#)

Ann Arbor, U.S.A

Aug. 2024

- Developed and delivered lectures on generative AI to high school students.

SERVICE

Reviewer: NeurIPS, ICML, ICLR, CVPR, TMLR, IEEE J-STSP, CPAL, ICASSP